Hill Choy

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SUMMARY

Software and Computer Vision Engineer experienced in deploying end-to-end AI pipelines and building robust backend systems. Adept at translating complex Edge AI concepts into practical solutions, utilizing novel technologies to solve real-life challenges and automate physical workflows for commercial clients and local authorities.

EDUCATION

University College London (UCL)
MSc Computer Graphics, Vision and Imaging — *Expected Graduation: September 2026*
- **Highlighted Coursework**: Machine Vision, ML for Visual Computing, Inverse Problems, Robot Vision.

Hong Kong University of Science and Technology (HKUST)
BEng in Computer Engineering (***Second Class Honors, Division 1***) — *Graduation: July 2024*
- **Highlighted Coursework**: Intro to NLP, Computer & Communication Networks.

IELTS Academic
8.0 (*Reading: 8.5, Listening: 9.0, Writing: 7.0, Speaking: 7.0*)

RELEVANT EXPERIENCE

Hyperspectral Image Analysis for Safe-Fabric Segmentation — **United Kingdom**
UCL, MSc Industry Dissertation Project (*Partner:* ***Zori Tex***) — *Ongoing*
- Collaborating with industry partner Zori Tex to research and develop an automated computer vision pipeline for textile waste sorting utilizing hyperspectral imaging.
- Exploring PCA and shallow networks to condense high-dimensional data for memory-efficient CNN processing, alongside a weakly supervised Autoencoder approach to handle uneven real-world illumination.

Research Assistant — **Hong Kong**
Prof. Shueng-Han Gary Chan, CSE, HKUST — *08/2024 - 06/2025*
- **K11 MUSEA Cars and Pedestrian Flow Monitoring System**
  - Engineered an end-to-end analytics system tracking 60,000+ daily visitors, utilizing OpenCV and a PyTorch-based YOLO model for real-time local flow counting on edge platforms.
  - Developed a multithreaded Flask backend utilizing Redis for real-time data synchronization and SQLite/ Pandas for robust storage, powering a web dashboard to optimize mall operations.
- **Privacy-Preserving Elderly Monitoring with a Novel Edge AI Camera**
  - Contributed to the development of a privacy-preserving Edge AI fall-detection system deployed on an NPU-accelerated OrangePi, utilizing Human Pose Estimation for real-time patient monitoring.
  - Developed rule-based post-processing algorithms to minimize false alarms without compromising true positive cases, achieving 98% accuracy and securing the Silver Award at the Hong Kong ICT Awards 2024.
- **Smart Traffic Management System (Kung Tong, Hong Kong)**
  - Trained and deployed YOLO models on a dual-RTX 4090 server to process live streams from 33 cameras across 11 major junctions, monitoring 20,000+ daily vehicles per junction.
  - Engineered automated client scripts using Python requests to securely transmit vehicle counts and yellow-box stop detections to a centralized API, handling Keycloak OAuth2 token authentication.

Advanced Video Analytics and Edge AI for Smart Carpark Systems — **Hong Kong**
HKUST, Bachelor Degree Final Year Project (*Supervisor:* ***Prof. Shueng-Han Gary Chan, CSE, HKUST***) — *09/2023 - 04/2024*
- Designed an Edge AI smart car park system to track real-time space availability, utilizing OpenCV for video frame grabbing and preprocessing to handle complex feeds from custom low-light and wide-angle cameras.
- Deployed a PyTorch-based YOLOv8 model on a Raspberry Pi by integrating a MobileNetV3 backbone, converting to ONNX format, and utilizing Ultralytics for custom dataset annotation, halving inference latency and reducing parameters by 30% while maintaining baseline accuracy.

Intern — **Hong Kong**
Hong Kong Telecom (HKT), Integrated Service Provision Center — *06/2023 - 04/2025*
- Engineered a Python GUI executable to replace a legacy system and eliminate manual checks, automating both Excel and physical report generation for phone line data.
- Bypassed the lack of a native API using a custom web-scraping pipeline, dynamically extracting interface records to synchronize data with a MySQL database.